Introduction to Galileo

Europe's Global Navigation Satellite System — Tutorial Notes

Preparation guide for the BIP26 Galileo Quiz | TU Dublin Tallaght Campus | June 2026

How to use these notes

These notes cover the key topics in the Galileo quiz — work through each section before attempting the quiz.
Questions marked 🔍 suggest a web search for the most current information — the answer may have changed since these notes were written.
A glossary of key terms and a list of reference links appear at the end of the document.

1. What is GNSS?

GNSS stands for Global Navigation Satellite System. It is the general term for any satellite-based system that provides positioning, navigation, and timing (PNT) services anywhere on Earth. GPS is one example of a GNSS — Galileo is another.

A GNSS system works by having a constellation of satellites in orbit broadcasting precise timing signals. A receiver on the ground listens to signals from multiple satellites simultaneously and uses the differences in signal arrival times to calculate its own position. This technique is called trilateration.

At least four satellites must be visible for a full 3D position fix (latitude, longitude, and altitude). With more satellites visible, accuracy improves.

GNSS System	Operator	Satellites (approx.)	Status
GPS	United States (DoD)	31	Fully operational
GLONASS	Russia	24	Fully operational
Galileo	European Union / ESA	~30	Fully operational
BeiDou	China	46+	Fully operational
QZSS	Japan	7	Regional (Asia-Pacific)
NavIC	India	7	Regional (South Asia)

2. Galileo — History and Context

Galileo is Europe's own civil global navigation satellite system, developed independently of the US GPS system. The key motivations for Europe developing its own system were:

- Independence — GPS is controlled by the US military and could theoretically be degraded or switched off for non-US users
- Civilian control — Galileo is under civilian governance, not military
- Higher accuracy — Galileo was designed from the outset with higher accuracy for civilian users
- Commercial and economic interest — Europe wanted to participate in the growing GNSS market

Milestone	Year
Galileo programme agreed by EU and ESA	2003
First Galileo test satellite (GIOVE-A) launched	2005
First operational satellites launched	2011
Initial Services declared (first usable signal)	2016
Full Operational Capability (FOC) achieved	2019
High Accuracy Service (HAS) activated	2023
OSNMA authentication service activated	2023

🔍 **Web search suggestion**

Search for 'Galileo satellite system status 2026' to find the current number of operational satellites and any recent service updates.

3. The Galileo Constellation

Galileo satellites orbit in Medium Earth Orbit (MEO) at an altitude of approximately 23,222 km. This is much higher than the International Space Station (~400 km) but much lower than geostationary satellites (~36,000 km).

- Orbital planes: 3 (equally spaced around the Earth)
- Satellites per plane: 10 (planned) — approximately 9-10 operational at any time
- Total operational satellites: approximately 28-30
- Orbital inclination: 56 degrees (tilted relative to the equator)
- Orbital period: approximately 14 hours 4 minutes

The MEO orbit means each satellite is visible from a given location on Earth for several hours at a time, and the constellation provides continuous global coverage.

Each Galileo satellite carries extremely precise atomic clocks:

- Passive Hydrogen Maser (PHM) — accuracy better than 1 nanosecond per day
- Rubidium Atomic Frequency Standard (RAFS) — backup clock

These clocks are essential because a timing error of just 1 nanosecond causes a position error of approximately 30 centimetres.

4. Galileo Services

Galileo provides several distinct services, each designed for different user groups:

Service	Acronym	Who Can Use It	Key Feature
Open Service	OS	Anyone — free	Standard positioning, dual-frequency
High Accuracy Service	HAS	Anyone — free	Sub-metre accuracy via corrections
Public Regulated Service	PRS	Government/authorised only	Encrypted, resistant to jamming
Search and Rescue Service	SAR	Emergency services	Detects distress beacons, sends acknowledgement
Commercial Authentication Service	CAS	Commercial users	Authentication for commercial applications

The Open Service is what most users (smartphones, car navigation, handheld GPS) receive. It is free, available globally, and provides positioning accuracy of approximately 1 metre horizontally when using both Galileo frequencies (E1 and E5).

The High Accuracy Service (HAS) broadcasts correction data on the E6 signal that allows compatible receivers to achieve sub-20 cm positioning accuracy — comparable to expensive professional RTK systems but free of charge.

5. OSNMA — Navigation Message Authentication

OSNMA stands for Open Service Navigation Message Authentication. It is one of Galileo's most significant technical innovations and directly relevant to BIP26.

The problem OSNMA solves: a malicious actor could broadcast fake GPS or GNSS signals (called spoofing) to make a receiver calculate an incorrect position. This is a serious security concern for critical infrastructure, autonomous vehicles, and precision timing systems.

OSNMA allows a receiver to verify that the navigation messages it is receiving genuinely come from authentic Galileo satellites and have not been tampered with.

5.1 How OSNMA Works

OSNMA uses a cryptographic protocol called TESLA (Timed Efficient Stream Loss-tolerant Authentication). The key concepts are:

- Galileo satellites broadcast authentication keys and digital signatures alongside the navigation data on the E1-B signal
- The receiver accumulates these keys and signatures over time
- Once a key is disclosed (after a short delay), the receiver can verify that all previous data signed with that key was authentic
- The root of trust is a public key pre-installed in receiver firmware — this allows the chain of authentication to be verified without any external connection

Key point for the quiz

OSNMA uses the TESLA protocol — Timed Efficient Stream Loss-tolerant Authentication.
OSNMA authenticates navigation messages (not just the signal). It tells you the data came from a real Galileo satellite.
OSNMA is broadcast on the Galileo E1-B signal frequency (1575.42 MHz).
OSNMA does not improve positioning accuracy — it improves trustworthiness.

Web search suggestion

Search for 'Galileo OSNMA status 2026' or 'ESA OSNMA receiver support' to find which GNSS receivers currently support OSNMA authentication.

6. Galileo Signals and Frequencies

Signal	Frequency	Service	Key Use
E1	1575.42 MHz	Open Service + PRS + OSNMA	Main civilian signal, authentication
E5a	1176.45 MHz	Open Service	Dual-freq accuracy improvement
E5b	1207.14 MHz	Open Service	Dual-freq accuracy improvement
E5 (AltBOC)	1191.795 MHz	Open Service	Combined E5a+E5b, highest accuracy
E6	1278.75 MHz	HAS + PRS	High accuracy corrections, gov. service

Using two frequencies (dual-frequency reception) allows a receiver to measure and correct for ionospheric delay — one of the largest sources of GNSS positioning error. This is why Galileo's dual-frequency Open Service achieves better accuracy than single-frequency GPS.

7. Galileo Ground Infrastructure

The ground segment of Galileo consists of:

- Galileo Control Centres (GCC) — two centres in Oberpfaffenhofen, Germany and Fucino, Italy. These monitor and control the entire satellite constellation.
- Galileo Sensor Stations (GSS) — a worldwide network of monitoring stations that track satellite signals and feed data back to the control centres
- Telemetry, Tracking and Command (TT&C) stations — communicate directly with satellites for orbit maintenance and clock corrections
- Up-link stations — broadcast navigation data and corrections up to the satellites

Web search suggestion

Search for 'European GNSS Agency EUSPA' or visit euspa.europa.eu for current information about Galileo operations and the agency responsible for it.

8. Galileo vs GPS — Key Differences

Aspect	Galileo	GPS
Operator	European Union / ESA (civilian)	US Department of Defense (military)
Control	Civilian — always available	Military — can be restricted
Accuracy (standard)	~1 m (dual-frequency)	~3-5 m (single frequency L1)

Aspect	Galileo	GPS
High accuracy	HAS: sub-20 cm, free	RTK: sub-cm but requires subscription
Authentication	OSNMA (free, built-in)	Not available on civilian signals
Atomic clocks	PHM + Rubidium	Caesium + Rubidium
First signal	2016	1978 (test), 1993 (operational)

9. Questions Requiring Web Searches

The following quiz questions may have answers that change over time. Use a web search to find the most current answer:

- Current number of operational Galileo satellites — search 'Galileo constellation status 2026'
- Current OSNMA receiver support — search 'GNSS receivers OSNMA support 2026'
- Current HAS accuracy figures — search 'Galileo HAS accuracy 2026'
- Cospas-Sarsat and Galileo SAR service — search 'Galileo search and rescue Cospas-Sarsat'

10. Glossary

Term	Definition
GNSS	Global Navigation Satellite System — any satellite-based positioning system
GPS	Global Positioning System — the US GNSS constellation
Galileo	Europe's GNSS constellation, civilian-controlled
OSNMA	Open Service Navigation Message Authentication — Galileo's anti-spoofing feature
TESLA	Timed Efficient Stream Loss-tolerant Authentication — the cryptographic protocol used by OSNMA
MEO	Medium Earth Orbit — approximately 20,000-23,000 km altitude
PNT	Positioning, Navigation, and Timing — the three core GNSS services
HDOP	Horizontal Dilution of Precision — measure of horizontal positioning geometry quality
HAS	High Accuracy Service — Galileo's free sub-metre correction service
PRS	Public Regulated Service — Galileo's encrypted government service
Spoofing	Broadcasting fake GNSS signals to deceive a receiver into a wrong position
Trilateration	Calculating position from distance measurements to multiple known points
Atomic clock	Extremely precise timekeeping device using atomic resonance — essential for GNSS
Dual-frequency	Using two signal frequencies to correct for ionospheric delay
E1	Galileo's main civilian signal frequency at 1575.42 MHz
E6	Galileo signal used for HAS corrections and PRS
EUSPA	European Union Agency for the Space Programme — operates Galileo
ESA	European Space Agency — developed and launched Galileo

11. Reference Web Links

The following links are recommended for quiz preparation:

- European Commission — Galileo introduction: https://defence-industry-space.ec.europa.eu
- Wikipedia — Galileo satellite navigation: https://en.wikipedia.org/wiki/Galileo_(satellite_navigation)
- EUSPA — Galileo services: https://www.euspa.europa.eu/european-space/galileo
- ESA — Galileo overview: https://www.esa.int/Applications/Navigation/Galileo
- OSNMA information: https://www.gsc-europa.eu/galileo/services/osnma
- Galileo service status: https://www.gsc-europa.eu/galileo-services/galileo-service-status
- Cospas-Sarsat SAR: https://www.cospas-sarsat.int